

Chemistry — Fraction Balancing Drill

Unit 6.1 · Combustion equations with odd O counts

Name: _____

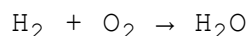
Block: _____

Date: _____

~20 minutes. Show all work. Balance one element at a time and save O₂ for last.

WORKED EXAMPLE — THE FRACTION MOVE

When balancing combustion, save O₂ for last. If you need an odd number of O atoms, write a fractional O₂ coefficient (like ½ or 5/2), then multiply every coefficient by the denominator to clear the fraction.

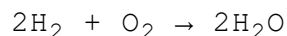


Step 1. Balance H → 1 H₂O gives 2 H ✓

Step 2. Balance O → right side has 1 O, but O₂ supplies 2 → write ½ O₂:



Step 3. Multiply every coefficient × 2 to clear the fraction:



Check: H = 4 = 4 ✓ O = 2 = 2 ✓

Your Turn

Q1. Practice — same move

5 pts

Balance the equation below using the fraction-then-multiply technique.

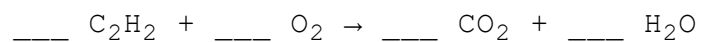


Balanced equation: _____

Q2. Build up

7 pts

Balance the combustion of acetylene. Save O₂ for last.

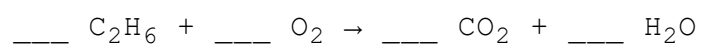


Balanced equation: _____

Q3. Full combustion

8 pts

Balance the combustion of ethane. Save O₂ for last.



Balanced equation: _____